

Jay Yadav

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jayyadaveng.github.io/portfolio/

WORK EXPERIENCE

- Textron Aviation: Airframe Design Engineer** Jun 2025 – Present
- Optimize aircraft structure, balancing stress, fatigue, MFG, M&P, and servicing requirements
 - Designed, drafted and released dozens of engineering packages in CATIA
 - Implemented productivity improvements through new designs involving multidisciplinary coordination
 - Modify and improve existing designs, and design structural components for novel aircraft
- Textron Aviation: Advanced Design Engineering Intern** May 2024 – Aug 2024
- Designed a baggage compartment mockup in CATIA V6 to test configurations for new aircraft
 - Researched emerging aerospace technology to develop novel aircraft systems
- Textron Aviation: Customer Support Engineering Intern** May 2023 – Aug 2023
- Created a technical troubleshooting system for the Cessna Citation Latitude
 - Updated and repaired troubleshooting systems for various Citation jets
- University of Arkansas: Mechanical Engineering REU Research Assistant** May 2022 – Aug 2022
- Designed in SOLIDWORKS and manufactured a lubricant spray coating system
 - Prepared and analyzed lubricants using 3D Laser and Scanning Electron Microscopy
 - Developed a graphite solid lubricant that reduces friction by 50%
 - Mentored by Engineering Dean in weekly leadership and communication sessions
- University of Texas at Austin: Electrical Engineering Research Assistant** Summer 2020, Summer 2021
- Researched in the Mid-IR Photonics Lab to develop novel infrared detectors
 - Expanded RCWA simulations in MATLAB to improve IR propagation modelling
 - Automated Python data collection and analysis to reduce dark current in IR detectors
- Columbia Machine Shop: Machining Superuser** May 2023 – May 2025
- Supervised student machine shop with 3D Printers and CNC mills and lathes

PROJECTS

- Columbia University Airplane Club: Member and Chief Design Engineer** Aug 2022 – May 2025
- Led aircraft CAD in SOLIDWORKS and analysis in Ansys for the Design/Build/Fly competition
 - Implemented rapid prototypes, parametric modelling, rigorous GD&T and CFD analysis
 - Oversaw the 2024 design to Columbia's most successful placement in 10 years
- Columbia Space Initiative Rocketry Team: Airframe Member** Aug 2022 – May 2025
- Optimized rocket nose and body with Ansys FEA and CFD for the Spaceport America Cup
 - Designed and manufactured composite carbon fiber rocket fins with layups and a CNC waterjet

EDUCATION

- Columbia University: Master of Science in Mechanical Engineering** Aug 2023 – May 2025
- GPA: 3.9/4.0
 - Selected Coursework: Advanced Fluid Mechanics, Advanced Thermodynamics, Robot Machine Learning
- Columbia University: Bachelor of Science in Mechanical Engineering** Aug 2022 – May 2024
- GPA: 4.1/4.0; Summa Cum Laude
 - Selected Coursework: Aerodynamics, Machine Design, Machining, Advanced Mfg, Aerospace Human Factors Engineering
- Grinnell College: Bachelor of Arts in Physics** Aug 2019 – May 2022
- GPA: 3.9/4.0
 - Selected Coursework: Electronics, Electromagnetic Theory, Astrophysics

SKILLS

- **Design:** Autodesk Inventor, ANSYS Mechanical and Fluent, CATIA V5/V6, SOLIDWORKS (CAD and FEA)
- **Coding:** C++, MATLAB, Python, HTML
- **Manufacturing:** Additive Manufacturing, CNC Machining, Composite Layups, DFM, GD&T, Soldering
- **Achievements and Hobbies:** Student Pilot, Eagle Scout, Aviation Enthusiast, Guitar, Piano, Basketball, Tennis